Integrals involving Fourier basis functions

The below thorem is used everywhere.

Theorem 0.1 (Fourier Integral):

$$\int_0^{2\pi} e^{imx}\,\mathrm{d}x = \begin{cases} 2\pi (m=0) \\ 0 \quad (m \in \mathbb{Z} \setminus \{0\}) \end{cases} \tag{1}$$

Proof:

$$\int_0^{2\pi} e^{imx}\,\mathrm{d}x = \frac{e^{im \cdot 2\pi} - e^{im \cdot 0}}{im} \tag{2.1}$$

$$= \begin{cases} 2\pi (m=0) \\ 0 \quad (m \in \mathbb{Z} \setminus \{0\}) \end{cases} \tag{2.2}$$

□

- Integral equations coming from common partial differential equations in $\mathbb{S}^1$ ($\mathbb{R}^2$) often involve singular integrals, derivative of integrals.
- To compute these integrals, exact values of integrals of Fourier basis functions are needed.
- To deal with the singularity while maintaining 2π-periodicity, we need a function which only has singularity at $0(2\pi)$ and is 2π-periodic.
  - $\csc^n t, \cot^n t, n \in \mathbb{N}$: singularities at π as well
  - $\csc^n \frac{t}{2}, n \notin 2\mathbb{N}$: not 2π-periodic
  - $\cot^n \frac{t}{2}, n \in \mathbb{N}$: ok
  - $\csc^{2n} \frac{t}{2}, n \in 2\mathbb{N}$: ok

Definition 0.1 (Cauchy Principle value on a circle): $\forall \alpha > 0. \forall t \in [0, 2\pi). \forall f \in C^{0,\alpha}.(0, 2\pi)$

$$⨍_0^{2\pi} f(x) \cot\left(\frac{x-t}{2}\right) \mathrm{d}x := \lim_{\varepsilon \to 0} \left(\int_\varepsilon^{2\pi - \varepsilon} f(x+t) \cot\left(\frac{x}{2}\right) \mathrm{d}x \right) \tag{3}$$

Definition 0.2 (Hadamard Finite-Part on a circle): $\forall \alpha > 0. \forall t \in [0, 2\pi). \forall n \in \mathbb{N} \setminus \{1\}. \forall f \in C^{n-1,\alpha}(0, 2\pi)$.

$$⨍_0^{2\pi} f(x) \cot^n\left(\frac{x-t}{2}\right) \mathrm{d}x := \lim_{\varepsilon \to 0} F_0(\varepsilon) \tag{4}$$

where

$$\forall \varepsilon \in (0, \pi). \int_\varepsilon^{2\pi - \varepsilon} f(x+t) \cot^n\left(\frac{x}{2}\right) \mathrm{d}x = F_0(\varepsilon) + F_1(\varepsilon) \tag{5.1}$$

$$\lim_{\varepsilon \to 0} F_0(\varepsilon) < \infty \tag{5.2}$$

$$\lim_{\varepsilon \to 0} \log\left(4 \sin^2\left(\frac{\varepsilon}{2}\right)\right) \frac{1}{F_1(\varepsilon)} < \infty \tag{5.3}$$

Theorem 0.2 (Hadamard Finite-Part on a circle): $\forall n \in \mathbb{N}. \forall f \in C^{n,\alpha}(\mathbb{R}/2\pi) \cap L^{1+}(\mathbb{R}/2\pi)$.Let

$$(T_n f)(t) := ⨍_0^{2\pi} f(x) \cot^n\left(\frac{x-t}{2}\right) \mathrm{d}x \tag{6}$$

then

$$(T_{n+1}f)(t) := -\frac{2}{n}\frac{\mathrm{d}}{\mathrm{d}t}(T_n f)(t) - (T_{n-1}f)(t) \tag{7}$$

where

$$(T_1 f)(t) := \fint_0^{2\pi} f(x)\cot\left(\frac{x-t}{2}\right)\mathrm{d}x \tag{8.1}$$

$$(T_0 f)(t) := \int_0^{2\pi} f(x)\,\mathrm{d}x \tag{8.2}$$

Theorem 0.3: $\forall n \in \mathbb{N}.\forall f \in C^{n,\alpha}(\mathbb{R}/2\pi) \cap L^{1+}(\mathbb{R}/2\pi)$.

$$\frac{\mathrm{d}}{\mathrm{d}t}\fint_0^{2\pi} f(x)\cot^n\left(\frac{x-t}{2}\right)\mathrm{d}x = \fint_0^{2\pi} f'(x)\cot^n\left(\frac{x-t}{2}\right)\mathrm{d}x \tag{9}$$

Proof:

$$\frac{\mathrm{d}}{\mathrm{d}t}\fint_0^{2\pi} f(x)\cot^n\left(\frac{x-t}{2}\right)\mathrm{d}x = \frac{\mathrm{d}}{\mathrm{d}t}\fint_0^{2\pi} f(x+t)\cot^n\left(\frac{x}{2}\right)\mathrm{d}x \tag{10.1}$$

$$= \frac{\mathrm{d}}{\mathrm{d}t}\lim_{\varepsilon\to 0} F(t,\varepsilon) \tag{10.2}$$

$$\underset{?}{=} \lim_{\varepsilon\to 0}\frac{\mathrm{d}}{\mathrm{d}t}F(t,\varepsilon) \tag{10.3}$$

$$= \fint_0^{2\pi} f'(x)\cot^n\left(\frac{x-t}{2}\right)\mathrm{d}x \tag{10.4}$$

□

Theorem 0.4:

$$\fint_0^{2\pi} f'(t)g(t)\,\mathrm{d}t \underset{?}{=} -\fint_0^{2\pi} f(t)g'(t)\,\mathrm{d}t \tag{11}$$

Theorem 0.5: Let

$$I_{m,n} = \fint_0^{2\pi} e^{imt}\cot^n\left(\frac{t}{2}\right)\mathrm{d}t \tag{12}$$

then following recurrence relation holds:

$$\forall m \in \mathbb{Z}.\forall n \in \mathbb{N}\setminus\{1\}.I_{m,n} = \frac{2im}{n-1}I_{m,n-1} - I_{m,n-2} \tag{13}$$

with initial values:

$$I_{m,0} = 0, \qquad I_{m,1} = 2\pi i\,\mathrm{sgn}(m) \quad (m \neq 0) \tag{14.1}$$

$$I_{0,0} = 2\pi, \qquad I_{0,1} = 0 \tag{14.2}$$

Proof:
- $I_{m,0}, m \in \mathbb{Z}$: Theorem 0.1.
- $I_{0,1} = 0$: Due to the asymmetry of integrand $\cot\frac{t}{2}$.
- $I_{m,1}, m \in \mathbb{Z}$: Follows [1, Lemma 8.23.].
  - $\cot\frac{t}{2} = \frac{\cos\frac{t}{2}}{\sin\frac{t}{2}} = i\frac{e^{\frac{it}{2}}+e^{-\frac{it}{2}}}{e^{\frac{it}{2}}-e^{-\frac{it}{2}}} = i\frac{e^{it}+1}{e^{it}-1}$

- $\therefore \forall m \in \mathbb{N}.(e^{imt}-1)\cot(\frac{t}{2}) = (e^{it}-1)\left(\sum_{j=0}^{m-1} e^{ijt}\right)\cot(\frac{t}{2}) = i(e^{it}+1)\sum_{j=0}^{m-1} e^{ijt}$.
- $\therefore \forall m \in \mathbb{N}.I_{m,1} - I_{0,1} = 2\pi i$ $\because$Integrating both hands and Theorem 0.1
- $\therefore \forall m \in \mathbb{N}.I_{m,1} = 2\pi i$
- $\forall m \in \mathbb{N}, I_{-m,1} = \overline{I_{m,1}} = -2\pi i$

- Recurrence relation: Due to Theorem 0.2 Theorem 0.3 and $\left(e^{imt}\right)' = ime^{imt}$.

□

Theorem 0.6: Let

$$J_{m,n} = \fint_0^{2\pi} e^{imt} \log\left(4\sin^2\left(\frac{t}{2}\right)\right)\cot^n\left(\frac{t}{2}\right) \mathrm{d}t \tag{15}$$

then following recurrence relation holds:

$$\forall m \in \mathbb{Z}.\forall n \in \mathbb{N} \setminus \{1\}.J_{m,n} = \frac{2im}{n-1} J_{m,n-1} - J_{m,n-2} + \frac{2}{n-1} I_{m,n} \tag{16}$$

with initial values:

$$J_{m,0} = -\frac{2\pi}{|m|}, \quad J_{m,1} = 2\pi i \operatorname{sgn}(m)\left(2H_{|m|} - \frac{1}{|m|}\right), H_m := \sum_{k=1}^{m} \frac{1}{k} \quad (m \neq 0) \tag{17.1}$$

$$J_{0,0} = 0, \qquad J_{0,1} = 0 \tag{17.2}$$

Proof:
- $J_{m,0}, m \in \mathbb{Z}$: Follows [1, Lemma 8.23.].
- $J_{0,1}$: Due to the asymmetry of integrand $\log(4\sin^2(\frac{t}{2}))\cot(\frac{t}{2})$.
- $J_{m,1}, m \in \mathbb{Z} \setminus \{0\}$:
  - $L(t) := \log(4\sin^2(\frac{t}{2})) = -\sum_{k\in\mathbb{Z}\setminus\{0\}} \frac{e^{ikt}}{|k|}$
  - $\forall m \in \mathbb{N}.J_{m,1} = \fint_0^{2\pi} e^{imt} L(t) L'(t)\,\mathrm{d}t = -\frac{im}{2}\int_0^{2\pi} e^{imt} L^2(t)\,\mathrm{d}t = -\frac{im}{2}\int_0^{2\pi} e^{imt}\left(\sum_{k\in\mathbb{Z}\setminus\{0\}} \frac{e^{ikt}}{|k|}\right)\left(\sum_{l\in\mathbb{Z}\setminus\{0\}} \frac{e^{ilt}}{|l|}\right)\mathrm{d}t = (2\pi)\sum_{k,l\in\mathbb{Z}\setminus\{0\},k+l+m=0} \frac{1}{|k||l|} = (2\pi)\sum_{k\in\mathbb{Z}\setminus\{0\},l\in\mathbb{Z},k+l+m=0} \frac{1}{|k||l|} = (2\pi)\sum_{k\in\mathbb{Z}\setminus\{0,-m\}} \frac{1}{|k||m+k|}$
  - $\forall m \in \mathbb{N}. \sum_{k\in\mathbb{Z}\setminus\{0,-m\}} \frac{1}{|k||m+k|} = \sum_{k=1}^{\infty} \frac{1}{k(m+k)} + \sum_{k=-m+1}^{-1} \frac{1}{|k||m+k|} + \sum_{k=-\infty}^{-m+1} \frac{1}{|k||m+k|} = \sum_{k=1}^{\infty} \frac{1}{k(m+k)} + \sum_{k=1}^{m-1} \frac{1}{k(m-k)} + \sum_{k=1}^{\infty} \frac{1}{k(m+k)} = \frac{2}{m}\sum_{k=1}^{\infty}\left(\frac{1}{k} - \frac{1}{m+k}\right) + \frac{1}{m}\sum_{k=1}^{m-1}\left(\frac{1}{k} + \frac{1}{m-k}\right) = \frac{2}{m}\left(\sum_{k=1}^{m} \frac{1}{k} + \sum_{k=1}^{m-1} \frac{1}{k}\right) = \frac{2}{m}\left(2H_m - \frac{1}{m}\right)$

□

Subspace U_N

Lemma 0.1: $\forall N' \in \mathbb{N}.t_j := \frac{2\pi j}{N'}.\forall m \in \mathbb{Z}.\forall t_{\text{start}} \in [0, 2\pi)$.

$$\frac{2\pi}{N'}\sum_{j=0}^{N'-1} e^{im(t_j - t_{\text{start}})} = \begin{cases} 2\pi (m \equiv 0 \bmod N') \\ 0 \quad (\text{otherwise}) \end{cases} \tag{18}$$

Definition 0.3: $U_N := \operatorname{span}\left(\left\{e^{imx} \mid m \in \mathbb{Z}, |m| < N\right\}\right)$

Theorem 0.7 (Trapezoidal Rule for U_N): $\forall N \in \mathbb{N}.N$-point trapezoidal rule is exact for U_N.

Proof: Lemma 0.1 □

Lemma 0.2 (Trapezoidal Rule for inner product for U_N): Let $N' := \dim U_N = 2N - 1$. Let $t_j := \frac{2\pi j}{N'}$ for $j = 0, ..., N' - 1$. $\forall f, g \in U_N$.

$$\int_0^{2\pi} f(t)g(t)\,\mathrm{d}t = \frac{2\pi}{N'} \sum_{j=0}^{N'-1} f(t_j) \tag{19}$$

Proof: Since, $f(t)g(t) \in U_{2N-1}$ Theorem 0.7 can be applied with $N' = 2N - 1$ points. □

Quadratures

Now we consider the way to express discrete Fourier expansion of a function in U_N using the values of the function at $t_{\text{start}}, t_{\text{start}} + \frac{2\pi}{N'}, t_{\text{start}} + \frac{4\pi}{N'}, \ldots$.

Lemma 0.3: $\forall N \in \mathbb{N}. \forall f \in U_N. N := 2N - 1. t_j := \frac{2\pi j}{N'}. \forall t_{\text{start}} \in [0, 2\pi)$.

$$f(x) = \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) \cdot \left(\frac{1}{N'} \sum_{|m|<N} e^{-im(t_j + t_{\text{start}})} e^{imx} \right) \tag{20}$$

Proof:

$$f(x) = \sum_{|m|<N} \int_0^{2\pi} f(t) \frac{e^{-imt}}{\sqrt{2\pi}}\,\mathrm{d}t \cdot \frac{e^{imx}}{\sqrt{2\pi}} \tag{21.1}$$

$$\underset{\because \text{ Lemma 0.2}}{=} \frac{1}{2\pi} \sum_{|m|<N} \frac{2\pi}{N'} \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) e^{-im(t_j + t)} \cdot e^{imx} \tag{21.2}$$

$$= \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) \cdot \left(\frac{1}{N'} \sum_{|m|<N} e^{-im(t_j + t_{\text{start}})} e^{imx} \right) \tag{21.3}$$

□

Combining the above lemma with the exact integral values of Fourier basis functions, we can exactly compute the integrals of functions in U_N multiplied by $\cot^n\left(\frac{t}{2}\right)$ or $\log\left(4\sin^2\left(\frac{t}{2}\right)\right)\cot^n\left(\frac{t}{2}\right)$, by using only the values of the function at $t_{\text{start}}, t_{\text{start}} + \frac{2\pi}{N'}, t_{\text{start}} + \frac{4\pi}{N'}, \ldots$.

Theorem 0.8 (Generalized Garrick–Wittich quadrature for U_N): $\forall n \in \mathbb{N}_0. \forall N \in \mathbb{N}. \forall f \in U_N. N' := 2N - 1. t_j := \frac{2\pi j}{N'}. \forall t_{\text{start}} \in [0, 2\pi)$.

$$\fint_0^{2\pi} f(t) \cot^n\left(\frac{t}{2}\right) \mathrm{d}t = \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) P_j^{N',n} \tag{22}$$

$$P_j^{N',n} := \frac{1}{N'} \sum_{|m|<N} I_{m,n} e^{-im(t_j + t_{\text{start}})} \tag{23}$$

Theorem 0.9 (Generalized Kussmaul–Martensen (Kress) quadrature for U_N): $\forall n \in \mathbb{N}_0. \forall N \in \mathbb{N}. \forall f \in U_N. N' := 2N - 1. t_j := \frac{2\pi j}{N'}. \forall t_{\text{start}} \in [0, 2\pi)$.

$$\fint_0^{2\pi} f(t) \log\left(4\sin^2\left(\frac{t}{2}\right)\right) \cot^n\left(\frac{t}{2}\right) \mathrm{d}t = \sum_{j=0}^{N'-1} f(t_j + t_{\text{start}}) Q_j^{N',n} \tag{24}$$

$$Q_j^{N',n} := \frac{1}{N'} \sum_{|m|<N} J_{m,n} e^{-im(t_j + t_{\text{start}})} \tag{25}$$

Error estimates

Integral Equations

Theorem 0.10:

$$\varphi(s) + \sum_{n=0}^{M} ⨍_0^{2\pi} K_n(s,t) \cot^n\left(\frac{s-t}{2}\right) \varphi(t)\,\mathrm{d}t + \sum_{n=0}^{M} ⨍_0^{2\pi} L_n(s,t) \log\left(4 \sin^2\left(\frac{s-t}{2}\right)\right) \cot^n\left(\frac{s-t}{2}\right) \varphi(t)\,\mathrm{d}t = f(s) \tag{26}$$

Bibliography

[1] R. Kress, *Linear Integral Equations*, vol. 82. in Applied Mathematical Sciences, vol. 82. New York, NY: Springer New York, 2014. doi: 10.1007/978-1-4614-9593-2.